Bayesian Optimization Report: Function 5

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.345	Low
2	0.351	Low
3	0.352	Low
4	0.357	Low
5	0.359	Low
6	0.361	Low
7	0.367	Low
8	0.373	Low
9	0.376	Low
10	0.381	Low

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	598.3013	425.0458	1267.5889
2	565.2443	405.5661	1090.7963
3	545.4087	387.7307	1075.5483
4	528.1880	375.4560	1016.1564
5	549.8752	379.4234	1045.6323
6	362.8444	34.1686	1058.7656
7	342.0612	36.9372	993.9693
8	335.8911	39.3455	1188.5099
9	310.9115	41.1415	1087.9302
10	322.9682	42.8214	970.3658

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.9390677 0.79878324 0.99793411 0.35071182]	6604.2888173453375	Matern(ν=0.5)	PORTFOLIO	0.345	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 26.726527675768}}

2	[0.98406242 0.40109357 0.91351569 0.71547901]	3567.4536650674395	Matern(ν=0.5)	PORTFOLIO	0.351	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 29.60207894070782}}
3	[0.91879927 0.96576617 0.97936606 0.22980351]	7076.80603959847	Matern(ν=0.5)	PORTFOLIO	0.352	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 29.697643659854997}}
4	[0.99890542 0.60479726 0.98839741 0.71974768]	5808.1280626220305	Matern(ν=0.5)	PORTFOLIO	0.357	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 32.31693534926363}}
5	[0.98790911 0.95930513 0.85889001 0.04059934]	4977.272171562603	Matern(ν=0.5)	PORTFOLIO	0.359	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 33.43710295002355}}
6	[1. 1. 1. 0.52214492]	8286.066621093983	Matern(ν=0.5)	PORTFOLIO	0.361	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 33.889206968623256}}
7	[1. 1. 1. 0.85310023]	8536.10261713755	Matern(ν=0.5)	PORTFOLIO	0.367	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 36.65329737489212}}
8	[1. 0.93816443 1. 1.]	8223.28651922745	Matern(ν=0.5)	PORTFOLIO	0.373	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 39.14785468661743}}
9	[1. 1. 1. 0.91389991]	8586.267887387137	Matern(ν=0.5)	PORTFOLIO	0.376	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 41.00222317381618}}
10	[1. 1. 1. 0.78514992]	8480.329120602279	Matern(ν=0.5)	PORTFOLIO	0.381	{'type': 'too_uncertain', 'severity': 0.5, 'details': {'avg_sigma': 42.80823904823014}}

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	2899.579531
EI	0.009027
PI	0.000063
THOMPSON	2384.169276
KG	0.009027
ENTROPY	inf

PORTFOLIO	2198.424517

Iteration 2

Method	Max Value
UCB	1793.625378
EI	0.000000
PI	0.000000
THOMPSON	1651.659969
KG	0.000000
ENTROPY	inf
PORTFOLIO	1821.057999

Iteration 3

Method	Max Value
UCB	2665.605916
EI	0.002891
PI	0.000029
THOMPSON	2604.894948
KG	0.002891
ENTROPY	inf
PORTFOLIO	2117.462010

Iteration 4

Method	Max Value
UCB	2245.240719
EI	0.000000
PI	0.000000
THOMPSON	2278.848327
KG	0.000000
ENTROPY	inf
PORTFOLIO	1681.508769

Iteration 5

Method	Max Value
UCB	2003.254982

EI	0.000000
PI	0.000000
THOMPSON	1702.680599
KG	0.000000
ENTROPY	inf
PORTFOLIO	1677.182212

Iteration 6

Method	Max Value
UCB	3171.844760
EI	13.680708
PI	0.173988
THOMPSON	3242.246003
KG	13.680708
ENTROPY	inf
PORTFOLIO	2352.823296

Iteration 7

Method	Max Value
UCB	3219.101786
EI	12.269268
PI	0.176016
THOMPSON	3209.826303
KG	12.269268
ENTROPY	inf
PORTFOLIO	2306.347311

Iteration 8

Method	Max Value
UCB	3259.626248
EI	9.507530
PI	0.177809
THOMPSON	3268.603394
KG	9.507530
ENTROPY	inf

PORTFOLIO	2297.231796

Iteration 9

Method	Max Value
UCB	3290.635866
EI	9.589288
PI	0.178576
THOMPSON	3304.041148
KG	9.589288
ENTROPY	inf
PORTFOLIO	2283.168116

Iteration 10

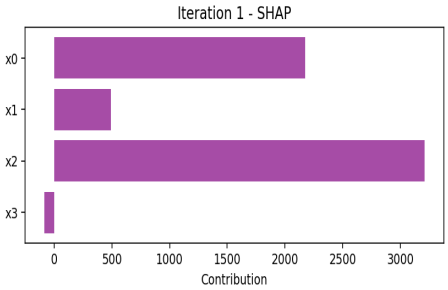
Method	Max Value
UCB	3324.898679
EI	7.244732
PI	0.180114
THOMPSON	3346.534065
KG	7.244732
ENTROPY	inf
PORTFOLIO	2292.110583

LOO MSE Summary

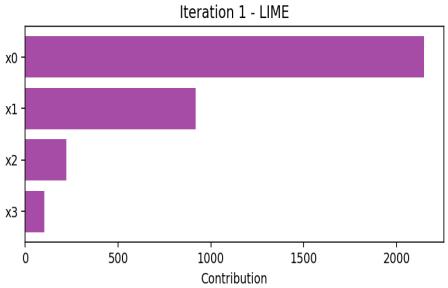
Iteration	LOO MSE
1	888220.543733
2	229628.050947
3	130710.165047
4	113690.795087
5	115100.322500
6	105004.832941
7	95094.981462
8	86672.444216
9	92188.034083
10	89643.949576

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



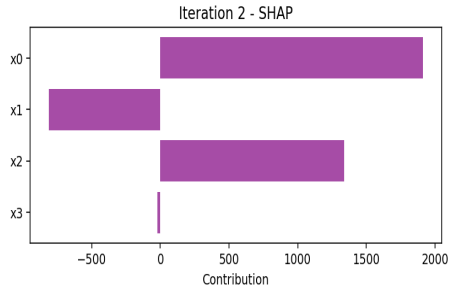
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	2174.11527
x1	491.43599
x2	3211.88754
x3	-85.12255

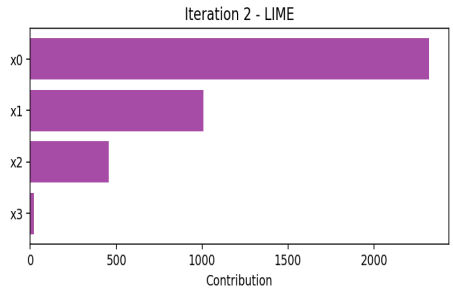
Iteration 1 - LIME Values

Feature	LIME Value
x0	2147.09786
x1	918.06467
x2	219.78868
x3	101.91066

Iteration 2 - SHAP



Iteration 2 - LIME



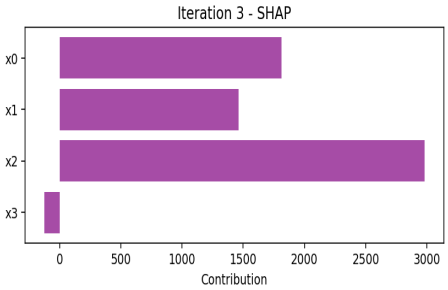
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	1910.64999
x1	-810.52439
x2	1339.05581
x3	-21.76793

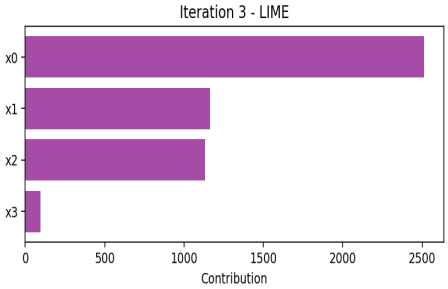
Iteration 2 - LIME Values

Feature	LIME Value
x0	2319.17554
x1	1008.61278
x2	456.95895
x3	21.50093

Iteration 3 - SHAP



Iteration 3 - LIME



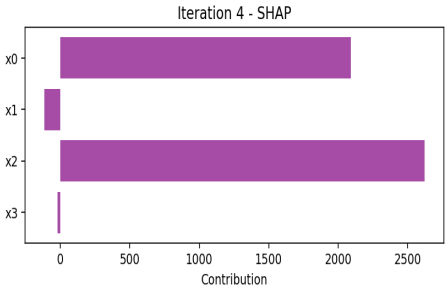
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	1811.98971
x1	1460.07846
x2	2983.39113
x3	-124.55796

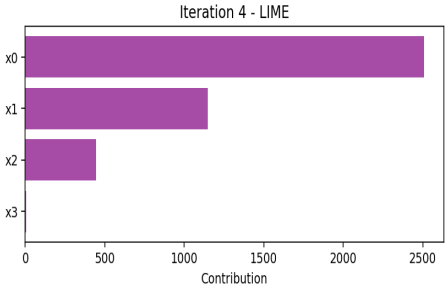
Iteration 3 - LIME Values

Feature	LIME Value
x0	2510.57444
x1	1162.21405
x2	1131.72687
x3	96.26068

Iteration 4 - SHAP



Iteration 4 - LIME



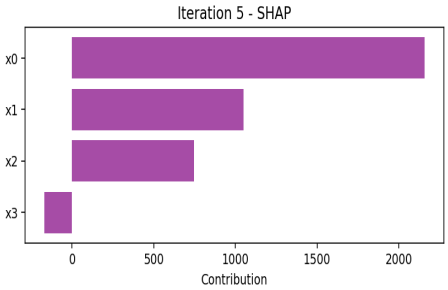
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	2092.50288
x1	-115.99707
x2	2624.14223
x3	-19.18348

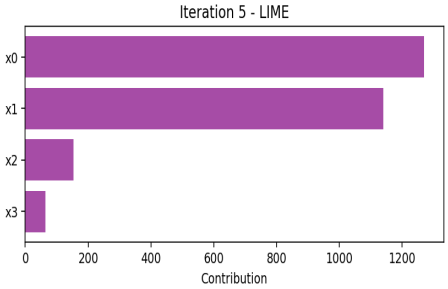
Iteration 4 - LIME Values

Feature	LIME Value
x0	2506.56687
x1	1146.44671
x2	446.31893
x3	6.60885

Iteration 5 - SHAP



Iteration 5 - LIME



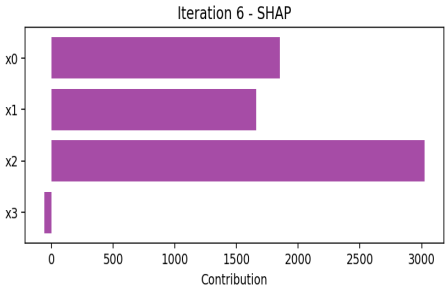
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	2159.63348
x1	1048.46776
x2	747.65964
x3	-169.69602

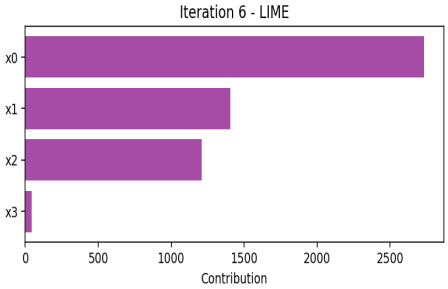
Iteration 5 - LIME Values

Feature	LIME Value
x0	1269.44082
x1	1140.15205
x2	153.51443
x3	64.81358

Iteration 6 - SHAP



Iteration 6 - LIME



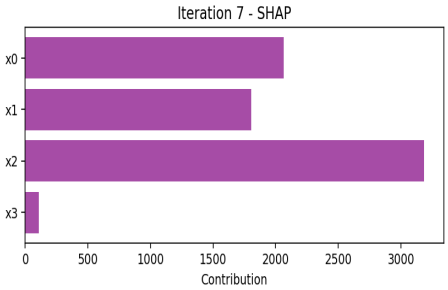
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	1852.93531
x1	1657.24307
x2	3025.88479
x3	-59.01443

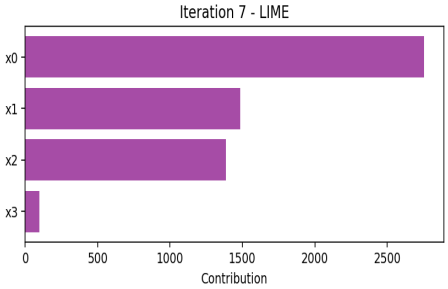
Iteration 6 - LIME Values

Feature	LIME Value
x0	2732.22724
x1	1404.36786
x2	1210.54196
x3	42.36994

Iteration 7 - SHAP



Iteration 7 - LIME



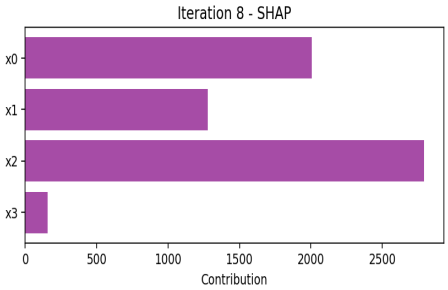
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	2065.64306
x1	1805.48545
x2	3183.41236
x3	107.10769

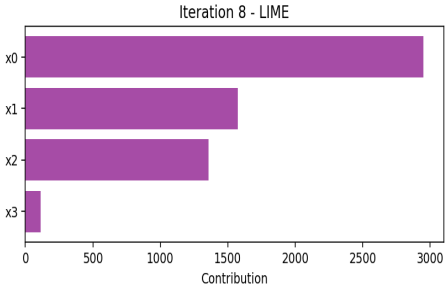
Iteration 7 - LIME Values

Feature	LIME Value
x0	2753.51053
x1	1485.79216
x2	1384.21714
x3	97.23520

Iteration 8 - SHAP



Iteration 8 - LIME



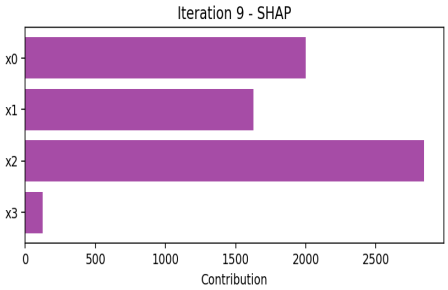
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	2005.61426
x1	1278.75997
x2	2791.00520
x3	154.94383

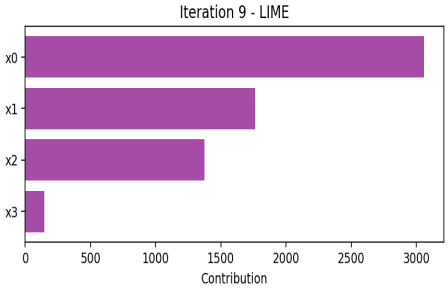
Iteration 8 - LIME Values

Feature	LIME Value
x0	2952.98857
x1	1575.06644
x2	1358.10958
x3	113.99174

Iteration 9 - SHAP



Iteration 9 - LIME



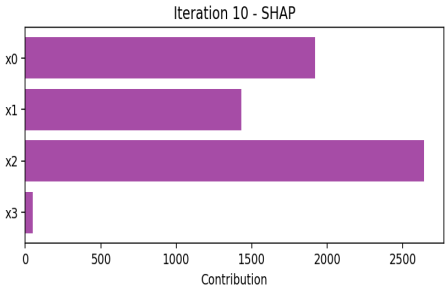
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	1998.81409
x1	1626.74905
x2	2842.67534
x3	125.07342

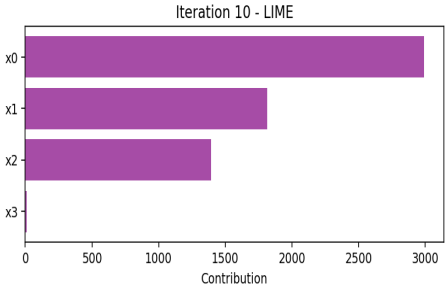
Iteration 9 - LIME Values

Feature	LIME Value
x0	3056.68660
x1	1764.17044
x2	1374.94144
x3	146.54882

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	1919.60589
x1	1430.96316
x2	2640.07366
x3	49.64447

Iteration 10 - LIME Values

Feature	LIME Value
x0	2990.01894
x1	1816.64142
x2	1391.94999
x3	12.44137